

ESV-Gap: Scope-Aware Evidence Control for Scientific Hypothesis Triage

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Abstract. Scientific agents can confuse missing graph relations or topical correspondence with evidence that a research question remains open. We develop an explicit scope contract for empirical hypotheses: one reviewed experiment must jointly satisfy the task, domain, conditions, and outcome requirements before it can close the specified question. The controller retains residual obligations, checks quoted provenance and recorded availability, and abstains when evidence or a declared search protocol is incomplete. Its correctness is conditional on supplied semantic annotations; it does not certify global novelty. A retrospective audit of 378 keyed records in four preserved corpora gives ten time-split evaluations and 64 non-independent heuristic future-control rows. At top 20, the earlier ESV-Gap candidate retriever attains macro proxy recall of 0.0139, versus 0 for confidence-ranked limitations and 0.0933 for BM25 reranking of the same extracted pool. All 39 ESV-Gap candidate rows originate in explicit limitations; the typed evidence-map branch yields none. An AI-assisted abstract audit does not independently confirm any of the five BM25 matches as a same-scope resolution. We implement the scope controller and pass 25 constructed policy tests, while preparing 188 blinded evidence pairs using raw-abstract retrieval for subsequent human assessment. These results motivate separating retrieval, evidence validity, and autonomous action; scientific efficacy of the new controller remains unmeasured.

Keywords: Scientific agents · Evidence scope · Hypothesis triage · Abstention · Literature-based discovery

1 Introduction

Literature-based discovery seeks connections between published findings [1]. Scientific AI assistants now generate questions, methods, and experimental plans using scholarly corpora and academic graphs [2]. Their autonomy creates a practical decision problem: which proposed questions warrant further formulation, which have already been answered within a specified scope, and which require more evidence? A plausible candidate is only the beginning of that decision.

A missing graph edge can reflect an unperformed experiment, an incomplete abstract, an extraction error, or inconsistent terminology. Likewise, a later paper mentioning a candidate’s terms may discuss an unresolved limitation, propose a conceptual framework, or evaluate a different task. Finding an action verb such

as “addresses” does not establish that the paper achieved a claimed outcome under the required conditions. Even a genuine measurement may be irrelevant to an underspecified question.

The original ESV-Gap architecture uses relation extraction, candidate generation, external checking, and an abstention policy. Its small IoT case study motivated a broader audit, but did not establish end-to-end discovery efficacy. In the preserved primary run, 133 raw signals were routed to zero automatically eligible candidates, five review candidates, and 128 rejected signals. These counts describe the stored gate output, not the five-case semantic reinterpretation used in the earlier manuscript. Keeping those stages separate avoids presenting a selected case proportion as population performance.

We investigate three questions. **RQ1:** How available and useful are candidates under a common temporal proxy evaluation? **RQ2:** Do proxy matches substantiate within-scope empirical resolution? **RQ3:** What explicit evidence obligations should govern an autonomous agent’s decision? We contribute (i) a fingerprinted four-corpus audit with matched-budget limitation rankings, (ii) a scope contract and controller that require a single reviewed experiment rather than a union of unrelated findings, and (iii) executable policy checks and a blinded annotation preparation workflow. The audit is retrospective; the new controller is a research implementation whose real-literature accuracy has not been measured.

2 Related Work and Contribution Boundary

ResearchAgent combines language models, academic-graph information, and iterative reviewing agents to develop ideas [2]. It includes human and model-based evaluation; we therefore avoid the blanket claim that prior scientific agents lack verification. Our narrower concern is the explicit evidence condition that permits a controller to suppress or retain a particular empirical question.

Müller-Bloch and Kranz’s research-gap framework already includes localization, characterization, verification, and presentation [3]. Its six types include contradictory evidence, knowledge void, action–knowledge conflict, methodological conflict, evaluation void, and theory application void. Our scope contract addresses empirical capability and evaluation questions; it is not a complete operationalization of that typology. Theoretical and methodological questions may require proofs or arguments rather than numerical witnesses.

SciFact studies scientific claim verification [4]; SciFact-Open studies retrieval in an open domain where evidence may support only a special case [5]. Context24 distinguishes stating a claim from empirically grounding it in methods, figures, or tables [6]. These findings motivate separate provenance and semantic-review checks. A source substring verifies where text came from, not whether it entails the annotated measurement or condition.

BM25 supplies a transparent lexical comparator [7]. Scientific document representations such as SPECTER and evaluation suites such as SciRepEval provide alternatives for a subsequent dense-retrieval comparison [8,9]; they have not been run here. Graph embeddings and communities can prioritize missing relation-

ships [10,11], but graph structure alone cannot establish experimental closure. Selective prediction motivates abstention [12]; our deterministic rule supplies neither a calibrated probability nor a population risk guarantee.

3 Scope Contract and Evidence Controller

3.1 Predeclared Empirical Question

Represent a question by

$$H = (\tau, d, C, R, t), \quad R = \{(m_j, \circ_j, b_j, u_j)\}_{j=1}^J, \quad (1)$$

where τ is the task, d the domain, C a nonempty set of required experimental conditions, R a nonempty set of outcome requirements, and t the evidence-availability cutoff. Each requirement names a metric m_j , a comparison $\circ_j \in \{\leq, \geq\}$, a finite threshold b_j , and a unit u_j . Requirements must be reviewed and fixed before assessing candidate witnesses. An underspecified phrase such as “Retrofitting–Security” fails this contract; the controller requests clarification rather than inventing a task or a success threshold.

For illustration only, a question might ask whether an IoT detector identifies unknown attacks on edge hardware while achieving F1 at least 0.90 and latency at most 10 ms. These are synthetic test thresholds, not validated deployment requirements or measured outcomes from our corpora. If latency is reported on a workstation, the hardware condition is unfulfilled. If one experiment reports F1 and another reports latency, they cannot be combined into a complete witness.

3.2 Reviewed Witness and Residual Obligations

An evidence record has a paper identifier, experiment identifier, recorded availability date, source text, and annotations for task, domain, conditions, and measurements. Each asserted facet carries an exact source span and a semantic-review flag. Metadata and experiment membership also require review: membership is the assertion that all annotated measurements concern the same experiment. The current implementation trusts those reviewed annotations; it does not automatically infer entailment or experiment identity from text.

Let $P(e)$ mean that identifiers, availability metadata, experiment membership, and all required quoted annotations pass these checks. Let $s_f(e, H)$ denote a satisfied facet. A within-scope witness is

$$W(e, H) = P(e) \wedge a(e) \leq t \wedge s_\tau(e, H) \wedge s_d(e, H) \wedge \bigwedge_{c \in C} s_c(e, H) \wedge \bigwedge_{r \in R} s_r(e, H). \quad (2)$$

The quantifier is $\exists e W(e, H)$: all obligations hold in one supplied experiment record. It is not $\bigwedge_f \exists e_f s_f(e_f, H)$, which could silently combine different studies, datasets, or hardware configurations. Duplicate records for the same paper–experiment key are flagged for review rather than merged.

Table 1. Scope-controller policy. Every branch records `novelty_established=false`. Closure is relative to the reviewed contract and annotations.

Condition, in precedence order	Disposition	Action
Invalid or unreviewed contract	Review required	Abstain and clarify
At least one complete witness	Closed within scope	Suppress candidate
Incomplete protocol or unchecked evidence	Review required	Abstain and escalate
Reviewed partial evidence	Partial evidence	Request residual evidence
No witness after completed protocol	Open for review	Abstain and escalate

For every non-witness, the audit records residual states: unreported, unreviewed or unquoted, different scope, incomparable units, or a failed threshold. Values with incompatible units are retained as unresolved rather than silently converted. Missing outcome data do not imply that a capability is impossible or scientifically novel. An annotation with a valid quotation but no semantic review is also unresolved.

3.3 Bounded Search and Action Policy

A search ledger declares required probe identifiers before recording results. Each required probe must have a provider, query, snapshot identifier, completed status, completed pagination, and completed semantic review. The ledger cutoff must equal the contract cutoff. Completion means the declared procedure completed; it is not exhaustive coverage of the literature, nor proof that a modern index reproduces a historical one.

Table 1 gives the implemented precedence. A complete witness can suppress an empirical absence candidate even if another probe failed: one valid existential witness suffices to close the specified question. Without such a witness, an incomplete search or unchecked evidence forces review. Reviewed evidence that satisfies only part of the contract produces residual obligations. A completed search with no witness yields only an open review queue.

Proposition 1 (Conditional witness integrity). *For a valid contract, the implemented controller returns closed within scope only if one supplied record satisfies every obligation in Eq. 2, relative to the trusted semantic and metadata annotations.*

Proof. The controller checks provenance and availability before evaluating a record. It evaluates every required facet and measurement in that record and marks it as a witness only when every state is satisfied. The closure branch requires at least one such witness. No branch merges record facets. Thus closure implies a complete individual witness. Semantic truth of the supplied annotations is an assumption, not a conclusion of this argument.

This is a property of the control rule. It neither establishes that source annotations are correct nor proves novelty when no witness is found. Vocabulary saturation, repeated limitation statements, and language-model confidence do not enter the witness rule as substitutes for empirical evidence.

4 Retrospective Audit Protocol

4.1 Preserved Corpora and Time Splits

The existing replay uses 378 keyed records: 150 IoT intrusion-detection papers, 143 microservice-security papers, 53 MongoDB-security papers, and 32 handwritten-mathematical-expression papers. Keys use DOI, identifier, or normalized title. The four corpora contain 3,188 extracted triples. Key-based deduplication does not guarantee that every preprint–publication pair was removed. These are preserved project corpora, not newly sampled or representative scientific fields.

Cutoffs are 2022, 2023, and 2024. A domain–cutoff evaluation requires at least ten records on both sides; ten qualify, of which nine contain positive heuristic controls. Pre-cutoff records and their extracted triples generate and rank candidates. Later triples generate proxy controls. A positive control can pair a limitation with a broad solution or evaluation relation, or instantiate a previously absent typed relation. These are silver controls, not expert-confirmed unresolved questions followed by experimentally established resolutions.

Candidate support identifiers are checked against the pre-cutoff set. All implemented checks passed. The split uses stored publication year, not independently reconstructed first-online dates. It does not establish historical availability of model training, prompts, embeddings, citation metadata, or search indexes. No online retrieval, new extraction, or end-to-end certification is included in this replay.

4.2 Matched-Budget Rankings and Proxy Metric

The three archived rankings are (i) ESV-Gap’s pre-certification candidate retriever, (ii) confidence ranking of direct **LACKS** triples without graph processing, and (iii) BM25 reranking of that same direct-limitation pool. The BM25 query is the domain label plus “limitation unresolved challenge,” with $k_1 = 1.2$, $b = 0.75$, and positive log-smoothed inverse document frequency. It was chosen post hoc. This ranking is extraction-dependent; it is not an independent raw-abstract gap-discovery baseline.

Every system inspects at most $k \in \{5, 10, 20\}$ candidates per evaluation under a common matching function. Matching uses normalized endpoints or capability token similarity of at least 0.62 with subject similarity of at least 0.25; an alternative capability threshold of 0.82 permits broader matches. Similarity combines token containment and Jaccard overlap. For nonempty positive controls H^+ ,

$$\text{ProxyRecall}@k = \frac{|\{h \in H^+ : \exists c \in C_{1:k}, c \sim h\}|}{|H^+|}. \quad (3)$$

Table 2. Per-cutoff audit. P_-/P_+ are pre/future records; H^+ are heuristic controls. Candidate counts and top-20 proxy recall describe the earlier retriever, not the new scope controller. A dash means no positive controls.

Domain/ t	P_-	P_+	H^+	ESV cand.	LACKS pool	ESV R20	BM25 R20
IoT/22	46	104	14	0	34	0	0
IoT/23	65	85	14	0	41	0	0
IoT/24	91	59	8	16	71	.125	.125
Microservice/22	47	96	14	5	50	0	.214
Microservice/23	79	64	8	6	84	0	0
Microservice/24	109	34	2	7	108	0	0
MongoDB/23	20	33	2	0	34	0	.500
MongoDB/24	27	26	1	1	44	0	0
Handwritten math/22	13	19	1	0	1	0	0
Handwritten math/23	18	14	0	4	7	–	–

Table 3. Matched-budget proxy correspondence across nine scored evaluations. The last column counts non-independent control rows. No entry estimates scientific novelty or scope-controller accuracy.

Ranking	R@5	R@10	R@20	Matches@20
ESV-Gap	0	0	0.0139	1/64
Confidence-LACKS	0	0	0	0/64
BM25-LACKS	0.0556	0.0694	0.0933	5/64

We average this value over the nine eligible evaluations with positive controls. The measure is correspondence with generated controls, not recall of genuine research gaps. Repeated cutoffs share papers and controls; the 64 control rows contain only 31 distinct control identifiers within domain. Neither fold averages nor row-level counts represent independent observations.

5 Audit Results

5.1 Candidate Yield and Proxy Correspondence

Table 2 reports the preserved replay. Four evaluations yield no ESV-Gap candidates, so their recall cannot be rescued by a better reranker. All 39 candidate rows arise from explicit limitations; the typed evidence-map branch yields zero. The direct-limitation pool contains 474 non-independent rows across cutoffs. These results give no positive evidence of graph-specific discovery value.

ESV-Gap matches one of eight IoT 2024 controls and none in the other eight scored evaluations; macro R20 is $0.125/9 = 0.0139$. BM25 matches five control rows across three evaluations. Two post-hoc BM25 query variants, replacing the generic suffix with “research gap limitation problem” or “limitation future work challenge,” each attain R20 of 0.0853. Their R5/R10 remain 0.0556/0.0694.

Table 4. Scope concerns in all five BM25 matches. These are AI-assisted abstract observations, not blinded human ground truth.

Candidate / later retained record	Evidential concern
IoT adversarial attacks / cybersecurity synthesis [13]	Attacks remain a challenge; no matched experiment is reported in the abstract.
Microservice retrofitting security / systematic mapping [14]	Surveys prior solutions; no new same-scope resolution experiment is specified.
Microservice retrofitting security / empirical IoT IDS [15]	DoS outcomes are reported, but the candidate does not declare that task or its required conditions.
API service-mesh security / Kubernetes overview [16]	No same-scope framework evaluation is reported in the abstract.
MongoDB platform security / retained security-analysis record [17]	Discusses measures; source identity and stored publication year remain unverified.

This checks wording sensitivity within the same extracted pool; it is not held-out model selection or a controlled estimate of graph benefit.

5.2 Why the Five Matches Are Not Closure Certificates

The five retained BM25 candidate-control matches were examined at abstract level with AI assistance. Four abstracts do not report a same-scope resolution experiment. A fifth reports quantitative DoS-detection results, but its matched candidate, “Retrofitting-Security,” specifies neither that task nor an experimental condition. Table 4 retains source traceability. No match has been independently confirmed as a complete within-scope witness; this is not equivalent to five verified false positives.

The MongoDB anomaly affects more than citation formatting: correcting its identity or date could change temporal assignment, controls, and scores. The frozen results retain the record for accounting and do not treat it as independently established future evidence. A provenance-corrected replay has not been performed. Full texts may contain experiments absent from these abstracts, so abstract-level insufficiency cannot establish the absence of a witness in the underlying publications.

As a post-hoc aggregate sensitivity check, omitting both MongoDB evaluations leaves seven scored evaluations and 61 control rows. Macro R20 becomes 0.0179 for ESV-Gap and 0.0485 for BM25-LACKS; confidence ranking remains zero. The ordering therefore persists without this domain, though the difference shrinks. This calculation omits an entire domain from saved aggregates; it neither repairs the disputed record nor validates the remaining controls.

6 Implemented Extension and Verification Status

6.1 Controller Checks

The new Python controller consumes a JSON contract, evidence records, and a search ledger, producing dispositions, residual traces, and an input SHA-256. It is an explicit opt-in component; it does not retroactively replace the earlier decision engine or alter historical runs. It accepts no vocabulary-saturation rule as evidence of experimental completeness. Boolean values are excluded from numeric measurements, nonfinite values never satisfy requirements, and incompatible units remain unresolved.

Twenty-five constructed tests pass. They check a complete individual witness, failure to combine measurements across experiments or papers, future-record exclusion, missing dates, unreviewed metadata and experiment membership, fabricated or unchecked quotes, wrong task/domain, missing or failed conditions, incompatible units, nonfinite values, failed outcome thresholds, underspecified or unreviewed contracts, missing probes, incomplete pagination, an empty completed search, witness precedence despite a failed probe, input immutability, duplicate experiments, and malformed metric names. These are software counterexamples and rule checks. Their pass rate is not an estimate of evidence-extraction accuracy, real-world hallucination suppression, or scientific-discovery success.

6.2 Extraction-Free Evidence Retrieval and Blinded Review Preparation

For each of the 39 archived candidate rows, a new local retriever forms a query from the pre-cutoff subject and capability and ranks raw titles plus abstracts using BM25. It excludes the candidate’s supporting paper identifiers and all records with stored years later than the cutoff; up to five nonzero-score records are retained. This produces 188 candidate–evidence pairs involving 97 distinct evidence-paper identifiers. They are preparation and workload counts, not efficacy metrics. Evidence ranking uses no extracted triples; candidate creation still does.

Two independently randomized CSV packets omit ranking scores and system judgments. Six annotation labels distinguish same-scope closure, partial evidence, supporting limitation, topical-only correspondence, insufficient evidence, and an underspecified candidate. Reviewers must supply quoted evidence and an experiment identifier for a closure label. All fields are initially blank; 39 contract templates deliberately leave task, conditions, and numerical requirements for review rather than generating unsupported success criteria. An agreement scorer rejects missing labels and overlapping reviewer identities, computes nominal Cohen’s κ , and lists disagreements. It does not mistake agreement for scientific correctness or system accuracy.

This workflow directly addresses the lack of independent semantic assessment, but the assessment has not yet occurred. It ranks evidence only within

Table 5. Distinct evaluation quantities for a subsequent labelled study. None is presently estimated for the new controller.

Quantity	Required denominator and interpretation
Witness retrieval recall	Adjudicated, within-scope witness documents retrieved / all known witness documents in the declared corpus.
Closure precision	Adjudicated valid closures / all controller closure decisions; report undefined if no closures.
Decision coverage	Cases receiving a closure decision / all valid scoped cases; report review workload separately.
Candidate quality	Expert-assessed specific and actionable questions / reviewed proposals, with unlabelled cases disclosed.
Operational cost	Queries, retrieved documents, semantic-review effort, latency, and model/API cost under common budgets.

the retained domain corpus and excludes origins by identifier, not by author group or complete preprint equivalence. Thus the workflow does not establish independent author corroboration or comprehensive external search. The controller was not evaluated on these 188 unannotated pairs, and none is used as a positive result for the new method.

7 Evaluation Needed for Scientific Efficacy

A prospective study should freeze scoped questions and requirements using only pre-cutoff information, then independently review evidence with system identity hidden. Questions should be pooled from direct limitations, an extraction-free proposal baseline, and graph proposals under equal candidate budgets. The same witness rules and retrieval budgets must apply to each arm. Dense and hybrid evidence retrievers can be added with frozen versions and query policies. A lexically strong ranking should not receive credit for a closure that lacks the required experimental scope.

At least two domain-qualified reviewers should first assess task/condition specificity, then annotate source-linked facets and experiment membership. Disagreements need adjudication by a third reviewer or a documented consensus process. Full text should be inspected for potential witnesses and abstracts marked insufficient when needed. The current 188-pair packet is a pilot instrument, not a sample size justified by a desired precision interval. Cases share candidates and papers; statistical analysis should account for that grouping.

Ablations should remove one requirement at a time: membership in one experiment, checked metadata, semantic review, or completion of the search. Studies under different conditions, reviews, conceptual proposals, and already resolved limitations should be explicit hard negatives. Retain graph complexity only if it adds expert-confirmed candidate utility. Keep open cases unresolved.

8 Limitations and Reproducibility

The audit uses four retained project corpora of unequal size and noisy extracted controls. Recorded years are a coarse historical proxy, and the MongoDB record has an unresolved provenance issue. Repeated folds reuse papers; no independence-based confidence intervals or significance tests are justified. The BM25 limitation comparator and query sensitivity were post hoc. The new raw-abstract evidence retriever is not an end-to-end gap-generation comparator and has no human relevance labels.

The scope controller depends on correct semantic, numerical, and metadata annotations. Exact quotations cannot prove entailment, and a reviewed flag does not independently verify its reviewer. A numeric contract is appropriate only when domain experts can defend meaningful metrics and thresholds. It may miss qualitative discoveries or theoretical contributions. Our conditional witness-integrity argument and synthetic tests supply control-rule evidence; they do not estimate real-literature accuracy or generalization.

Retained reports contain SHA-256 fingerprints of corpus/triple inputs, configuration, and replay code. Preparation reports additionally hash the four corpora and ten archived fold reports. The aggregate values in Tables 2–3 can be checked from those retained reports; full-pipeline reproduction additionally requires the original inputs and environment. Scripts, reports, packets, and source are retained locally. No public archival release, independent reproduction, redistributability of abstracts, or historically complete search snapshot is claimed.

9 Conclusion

The retained four-corpus audit exposes weak candidate availability and a gap between proxy correspondence and empirical resolution. BM25 reranking improves the proxy score within an extracted limitation pool, but none of its five matches is independently confirmed as a same-scope closure. We implement a scope controller that requires one reviewed experiment to meet every declared obligation and keeps absence claims in review. Twenty-five constructed checks validate policy behavior, and 188 blinded evidence pairs support a subsequent annotation study. The next evidential requirement is independent, source-linked evaluation of the new controller; its scientific efficacy remains an open empirical question.

AI assistance disclosure. Generative AI assisted writing, code development, and abstract-level inspection. Generated text and software tests are not treated as independent expert annotation or scientific evidence.

References

1. Swanson, D.R.: Fish oil, Raynaud’s syndrome, and undiscovered public knowledge. *Perspectives in Biology and Medicine* 30(1), 7–18 (1986)
2. Baek, J., Jauhar, S.K., Cucerzan, S., Hwang, S.J.: ResearchAgent: Iterative research idea generation over scientific literature with large language models. In: NAACL, pp. 6709–6738 (2025). <https://doi.org/10.18653/v1/2025.naacl-long.342>
3. Müller-Bloch, C., Kranz, J.: A framework for rigorously identifying research gaps in qualitative literature reviews. In: ICIS, Research Methods, paper 2 (2015). <https://aisel.aisnet.org/icis2015/proceedings/ResearchMethods/2/>
4. Wadden, D., et al.: Fact or fiction: Verifying scientific claims. In: EMNLP, pp. 7534–7550 (2020). <https://doi.org/10.18653/v1/2020.emnlp-main.609>
5. Wadden, D., et al.: SciFact-Open: Towards open-domain scientific claim verification. In: Findings of EMNLP, pp. 4719–4734 (2022). <https://doi.org/10.18653/v1/2022.findings-emnlp.347>
6. Chan, C.S.J., et al.: Overview of the Context24 shared task on contextualizing scientific claims. In: Workshop on Scholarly Document Processing, pp. 12–21 (2024). <https://aclanthology.org/2024.sdp-1.3/>
7. Robertson, S., Zaragoza, H.: The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval* 3(4), 333–389 (2009). <https://doi.org/10.1561/15000000019>
8. Cohan, A., et al.: SPECTER: Document-level representation learning using citation-informed transformers. In: ACL, pp. 2270–2282 (2020). <https://doi.org/10.18653/v1/2020.acl-main.207>
9. Singh, A., et al.: SciRepEval: A multi-format benchmark for scientific document representations. In: EMNLP, pp. 5548–5566 (2023). <https://doi.org/10.18653/v1/2023.emnlp-main.338>
10. Bordes, A., et al.: Translating embeddings for modeling multi-relational data. In: *Advances in Neural Information Processing Systems* 26 (2013)
11. Blondel, V.D., et al.: Fast unfolding of communities in large networks. *Journal of Statistical Mechanics*, P10008 (2008). <https://doi.org/10.1088/1742-5468/2008/10/P10008>
12. Geifman, Y., El-Yaniv, R.: Selective classification for deep neural networks. In: *Advances in Neural Information Processing Systems* 30 (2017)
13. Jayawardena, D., Perera, K.: AI-driven cybersecurity frameworks for real-time intrusion detection and threat intelligence. *Int. J. Comput. Sci. Inf. Syst.* 11(7), 60–77 (2026). <https://doi.org/10.55640/ijcsis/Volume11Issue07-02>
14. Rahaman, M.S., Islam, A., Cerny, T., Hutton, S.: Static-analysis-based solutions to security challenges in cloud-native systems: Systematic mapping study. *Sensors* 23(4), 1755 (2023). <https://doi.org/10.3390/s23041755>
15. Plazas Olaya, M.K., Vergara Tejada, J.A., Aedo Cobo, J.E.: Securing microservices-based IoT networks: Real-time anomaly detection using machine learning. *J. Comput. Networks Commun.*, 9281529 (2024). <https://doi.org/10.1155/2024/9281529>
16. Kampa, S.: Navigating the landscape of Kubernetes security threats and challenges. *J. Knowledge Learning Sci. Technol.* 3(4), 274–281 (2024). <https://doi.org/10.60087/jklst.v3.n4.p274>
17. Singh, P.K., et al.: An analysis and overview of MongoDB security. *SSRN Electronic Journal* (stored year 2024). <https://doi.org/10.2139/ssrn.4759892>. Retained metadata only; source identity and publication year unverified.